

Lecture 10 : IP subnetting

Introduction

In the vast landscape of computer networks, IP addresses serve as unique identifiers for every device. Traditionally, these addresses were categorized into classes (A, B, C), each with a predefined structure for network and host identification. However, as networks grew in complexity and the demand for IP addresses surged, the need for more flexible and efficient address management became paramount. This lecture delves into the concept of IP subnetting, a crucial technique that allows network administrators to divide a single IP network into smaller, more manageable sub-networks.

Subnetting not only enhances network organization and security but also plays a vital role in conserving the limited pool of IPv4 addresses. We will explore how IP masks are used to define these sub-networks and then examine two advanced methods for creating them efficiently: Variable Length Subnet Mask (VLSM) and Classless Inter-Domain Routing (CIDR). These techniques move beyond the rigid class-based system, offering unprecedented flexibility in network design and address allocation, which is essential for modern, scalable network infrastructures.

Core Concepts

Term	Definition	Significance
IP Subnetting	The process of dividing a larger IP network into smaller, distinct sub-networks, each with its own unique subnet ID.	Improves network management, enhances security by isolating traffic, and optimizes the use of IP address space.
IP Mask	A 32-bit number, typically represented in dotted decimal notation, used to differentiate the network portion of an IP address from the host portion.	Essential for routers and devices to determine which part of an IP address identifies the network and which identifies a specific host within that network.
Natural Mask	The default subnet mask associated with traditional Class A, B, or C IP	Represents the original, fixed hierarchy of IP address classes before custom subnetting was

Term	Definition	Significance
	addresses, defining their standard network and host bit divisions.	introduced, serving as a baseline for understanding IP structure.
Network Part	The initial bits of an IP address that identify the specific network to which a device belongs.	Crucial for routing data between different networks; all devices on the same network share the same network part.
Host Part	The remaining bits of an IP address that uniquely identify a specific device within a given network.	Allows individual devices to be addressed within their local network; all devices on a network must have a unique host part.
Subnet Part	Bits borrowed from the host portion of an IP address to create additional sub-networks within a larger network.	Introduces a third level of hierarchy (network, subnet, host), enabling finer-grained network segmentation and more efficient address allocation.
VLSM (Variable Length Subnet Mask)	A subnetting technique that permits the use of different subnet masks within the same IP network, creating subnets of varying sizes.	Maximizes IP address utilization by tailoring subnet sizes to the actual number of hosts required by different segments of an organization.
CIDR (Classless Inter-Domain Routing)	An IP addressing method that abandons the traditional Class A, B, C system, using a flexible prefix length (e.g., /24) to define network boundaries.	Reduces the size of routing tables, improves routing efficiency, and significantly conserves IP addresses by allowing networks of arbitrary sizes.

Detailed Analysis

1. The Foundation of IP Subnets: Natural Masks and Hierarchical Addressing

IP networks traditionally employ a two-level hierarchical addressing scheme, distinguishing between a network part and a host part. For instance, in a Class A, B, or C network, the initial bits define the network, while the remaining bits identify individual hosts. To introduce a third level of hierarchy—the sub-network—we utilize network masks.

A mask is a 32-bit sequence of ones and zeros. The leading ones correspond to the network (and subnet) portion, while the trailing zeros correspond to the host portion. When an IP address is bitwise ANDed with its mask, the result reveals the network address.

- **Natural Masks for Classful IP Addressing:**

- **Class A Network:** The first 8 bits represent the network, and the remaining 24 bits are for hosts. Its natural mask is `255.0.0.0` (eight 1s followed by twenty-four 0s).
- Example: For an IP address `10.0.0.20` in a Class A network, ANDing it with `255.0.0.0` yields `10.0.0.0`, which is the network number.
- **Class B Network:** The first 16 bits represent the network, and the remaining 16 bits are for hosts. Its natural mask is `255.255.0.0` (sixteen 1s followed by sixteen 0s).
- **Class C Network:** The first 24 bits represent the network, and the remaining 8 bits are for hosts. Its natural mask is `255.255.255.0` (twenty-four 1s followed by eight 0s).

These natural masks enforce a fixed division, but subnetting allows for more flexible, arbitrary masks.

2. Subnetting with Flexible Masks: Extending the Hierarchy

Subnetting involves borrowing bits from the host portion of an IP address to create a subnet portion. This effectively partitions the original host address space into multiple smaller subnets, each with its own range of host addresses. The network mask is then extended to include these borrowed bits, defining the new subnet boundaries.

- **Process of Subnetting:**

- Start with a given IP address and its natural mask.
- Identify the host portion of the address.

- Borrow a certain number of bits from the host portion to create the subnet ID.
- The remaining bits in the host portion are used for host identification within that specific subnet.
- The subnet mask is modified by setting the borrowed bits to '1'.
- **Example: Subnetting a Class A Network:**
 - Consider a Class A network `10.0.0.0` with a natural mask of `255.0.0.0`. This network has 24 bits available for hosts, allowing for $2^{24} - 2$ hosts.
 - If we use a subnet mask of `255.255.0.0`, we are effectively borrowing 8 bits from the host portion (the second octet) to create subnets.
 - The first 8 bits (`10`) remain the network ID. The next 8 bits (the second octet) now represent the subnet ID. The last 16 bits are for hosts within that subnet.
 - Illustration: For an IP address `10.5.0.20` with the subnet mask `255.255.0.0`, a bitwise AND operation yields `10.5.0.0`. Here, `10` is the network ID, and `5` is the subnet ID.
 - This configuration allows for $2^8 = 256$ possible subnets, with each subnet supporting $2^{16} - 2 = 65534$ hosts.
- **Example: Subnetting a Class B Network:**
 - A Class B IP address like `144.16.72.20` has a default mask of `255.255.0.0`. The network address is `144.16.0.0`.
 - If a custom subnet mask like `255.255.192.0` is used, it means 2 bits have been borrowed from the third octet (192 in binary is `11000000`).
 - ANDing `144.16.72.20` with `255.255.192.0` results in `144.16.64.0`. Here, `144.16` is the network ID, and `64` is the subnet ID within that network. This leaves 14 bits for hosts ($2^{14} - 2$ hosts).

3. Variable Length Subnet Mask (VLSM): Tailoring Subnet Sizes

VLSM is an advanced subnetting technique that allows for the creation of subnets with different sizes within the same network. This is particularly useful for organizations with varying departmental host requirements, as it significantly improves IP address utilization compared to fixed-size subnetting.

- **Problem with Fixed-Size Subnetting:** If an organization needs subnets for departments with 110, 45, and 50 computers, a fixed-size subnetting approach (e.g., dividing a Class C network into four equal subnets of 64 hosts each) would fail to accommodate the largest department (110 hosts) and waste addresses in smaller departments.

- **VLSM Solution:**

- **Step 1: Initial Division:** Start with the largest subnet requirement. For a Class C network (e.g., `192.203.17.0`), which supports 254 hosts, divide it into two subnets of 128 hosts each using a mask like `255.255.255.128`.
- Subnet 1: Host range 0-127.
- Subnet 2: Host range 128-255.
- **Step 2: Assign Largest Department:** Assign the first subnet (0-127) to the department requiring 110 hosts (D1).
- **Step 3: Further Subdivision:** Take the remaining subnet (128-255) and further divide it to accommodate the next largest requirements. Divide it into two subnets of 64 hosts each using a mask like `255.255.255.192` (192 in binary is `11000000`, borrowing 2 bits).
- Subnet 2.1: Host range 128-191.
- Subnet 2.2: Host range 192-255.
- **Step 4: Assign Remaining Departments:** Assign Subnet 2.1 (128-191) to D2 (45 hosts) and Subnet 2.2 (192-255) to D3 (50 hosts).
- This hierarchical division, using different masks at different levels, ensures that each department receives an appropriately sized subnet, minimizing wasted IP addresses. VLSM requires routers within the organization to support this capability.

4. Classless Inter-Domain Routing (CIDR): The Modern Addressing Standard

CIDR represents a fundamental shift from classful IP addressing, eliminating the rigid Class A, B, C boundaries. It provides a more flexible and efficient way to allocate IP addresses and route traffic, which is now the standard in modern internet routing.

- **CIDR Notation:** An IP address is followed by a slash and a number (M), like `144.16.192.57/18`.
- The number `M` indicates the number of bits in the network portion (prefix length).
- The remaining `32 - M` bits are used for the host portion.
- Example: `144.16.192.57/18` means the first 18 bits define the network, and the remaining 14 bits define the host.
- **Advantages of CIDR:**
- **Flexible Network Sizing:** Allows organizations to be assigned IP address blocks of any size, not just fixed Class A, B, or C blocks.

- **Reduced Routing Table Size:** Routers can aggregate multiple smaller networks into a single routing table entry, simplifying routing and reducing memory requirements.
- **IP Address Conservation:** Prevents the wasteful allocation of large classful blocks to organizations that only need a few addresses.
- **Rules for CIDR Blocks:**
 - **Power of 2:** The total number of addresses in any CIDR block must be a power of 2 (e.g., 2, 4, 8, 16, etc.), as it's determined by the number of host bits ($2^{\text{host_bits}}$).
 - **Divisible Starting Address:** The beginning address of a CIDR block must be evenly divisible by the total number of addresses in that block. This means the host portion of the starting address must always be all zeros.
 - **Example:** For a block of 16 addresses (4 host bits), the starting address must have its last 4 bits as 0000. An address like X.Y.Z.64 (where 64 in binary is 01000000) is valid as a starting address for a 16-address block because 64 is divisible by 16. An address like X.Y.Z.65 would not be valid.
- **Example: Analyzing a CIDR Block:**
 - Given 144.16.192.40/29 :
 - /29 means the first 29 bits are for the network, and $32 - 29 = 3$ bits are for hosts.
 - Number of addresses in the block: $2^3 = 8$.
 - To find the starting address, set the last 3 host bits to 000 . For 144.16.192.40 , the binary for 40 is 00101000 . Setting the last 3 bits to 000 gives 00101000 which is 40 . So the starting address is 144.16.192.40 .
 - To find the ending address, set the last 3 host bits to 111 . For 144.16.192.40 (00101000), setting the last 3 bits to 111 gives 00101111 , which is 47 . So the ending address is 144.16.192.47 .
 - The range of valid IP addresses in this block is 144.16.192.40 to 144.16.192.47 .
 - CIDR is a generalization of classful addressing; a Class A network is equivalent to /8 , Class B to /16 , and Class C to /24 . All modern routers support CIDR, making it the prevalent addressing scheme today.

Key Takeaways

- [] IP subnetting divides a large IP network into smaller sub-networks for better management and address utilization.

- [] IP masks are used to distinguish the network/subnet portion from the host portion of an IP address.
- [] Natural masks define the fixed network and host parts for traditional Class A, B, and C IP addresses.
- [] Subnetting introduces a third hierarchical level by borrowing bits from the host part to create a subnet ID.
- [] Bitwise ANDing an IP address with its subnet mask reveals the network or subnet address.
- [] Variable Length Subnet Mask (VLSM) allows for subnets of different sizes within the same network, optimizing address space.
- [] Classless Inter-Domain Routing (CIDR) replaces classful addressing with a flexible prefix length (/M) to define network boundaries.
- [] CIDR blocks must contain a number of addresses that is a power of 2, and their starting address must be divisible by the block size.
- [] CIDR helps conserve IP addresses and reduces routing table sizes, making it the standard for modern internet routing.
- [] All contemporary routers are designed to support CIDR for efficient network operations.

Conclusion

The journey from rigid classful IP addressing to the flexible paradigms of VLSM and CIDR reflects the continuous evolution of network design to meet the demands of an ever-expanding internet. Subnetting, at its core, provides a powerful mechanism for network administrators to segment their networks, enhancing security, improving performance, and simplifying management. VLSM takes this a step further by allowing for the creation of subnets tailored to specific organizational needs, thereby maximizing the efficient use of valuable IP address space.

CIDR, as the current standard, completely revolutionizes how IP addresses are allocated and routed. By decoupling network size from traditional classes, it offers unparalleled flexibility, reduces the burden on routing infrastructure, and ensures the sustainable growth of the internet. Understanding these concepts is not merely academic; it is fundamental for anyone involved in designing, implementing, or managing modern IP networks, enabling them to build robust, scalable, and efficient communication infrastructures.